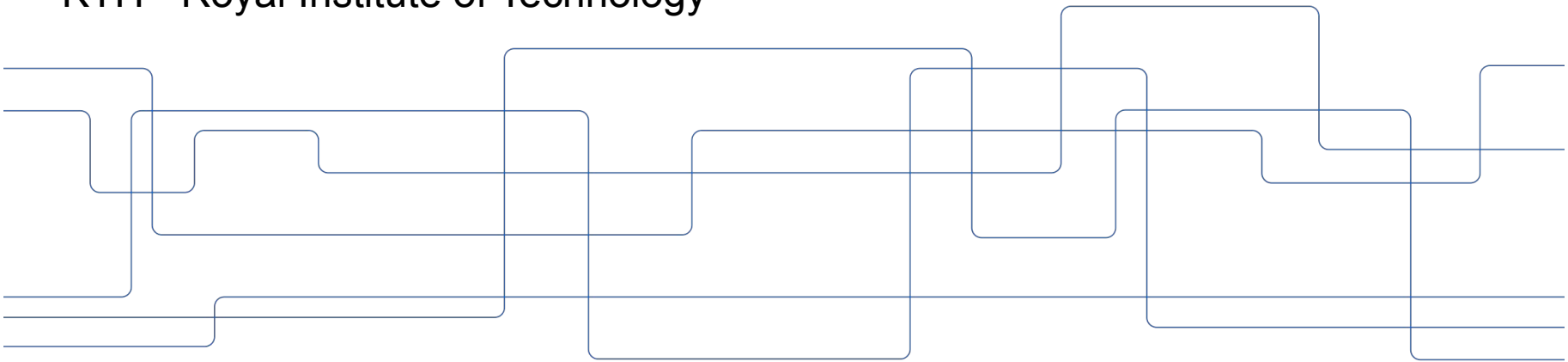
Computer lab- Molecular Simulation

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Presentation – Outline

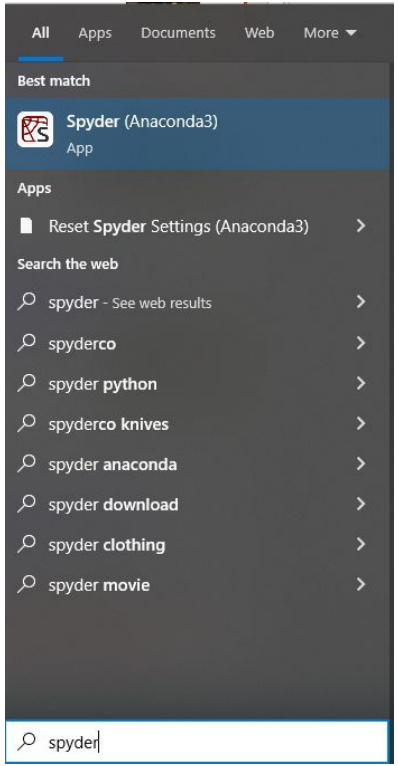
- General description of the Python code and simulation
- Using Spyder to run the simulations
- Report's description
- Report correction and evaluation

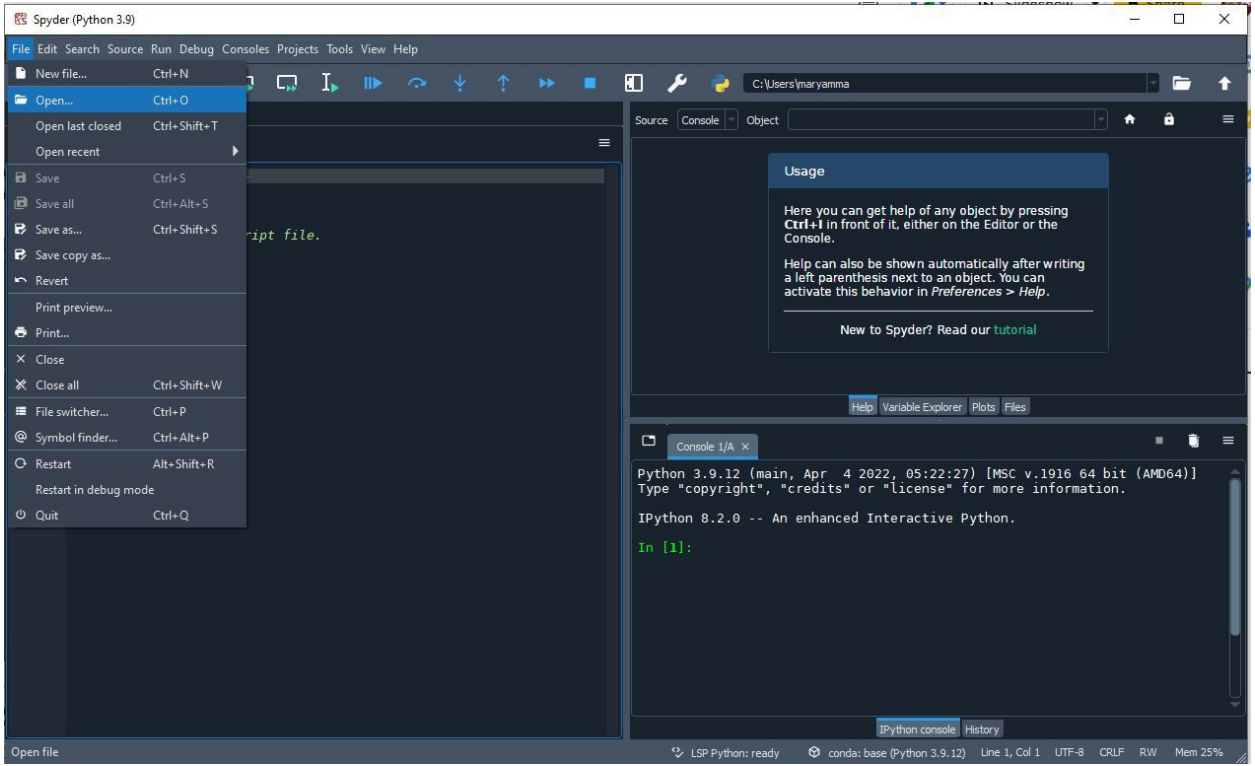
General description of the simulation

- Simulate the dynamic behavior of N particles of given mass in 2 dimensions at a given temperature T, for the specified duration integrating the Newton's equation of motion.

  - Download the python script md_exercise.py (on CANVAS)
  - Open the script using Spyder

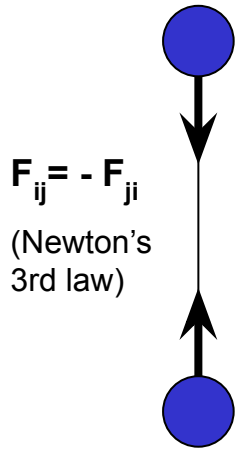
Open the script using Spyder





Description of the simulation and the Python script

- Define r(0) and v(0) for each atom
  - In the initial configuration the particles are at a distance of at least 0.35. The starting velocities are randomly generated according to the given temperature.
- Calculate the configuration of the system over time using Newton's equations of motion
  - F = ma
  - $F_{ij} = - F_{ji}$



How to calculate force

The particles interact with each other via van der Waals interactions, which we model using the Lennard-Jones potential:

$$V_{VDW}(r_{ij}) = 4\varepsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right]$$

The parameter ε is the well-depth and the parameter σ is the distance at which the potential is zero.

$$F_i = -\frac{\partial}{\partial r_i} V(r_1, r_2 \ldots\ldots\ldots r_N)$$

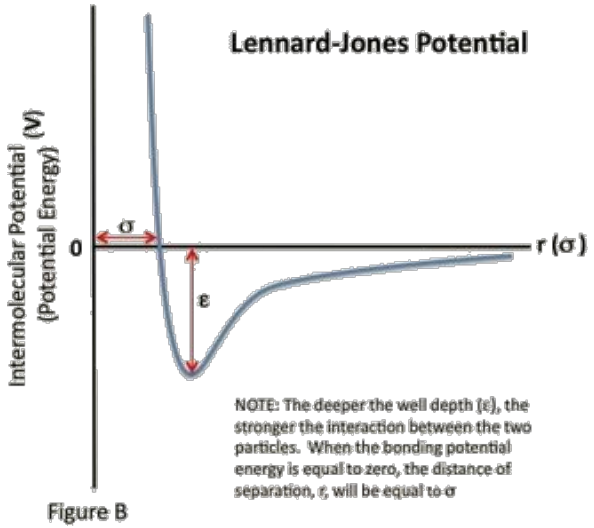


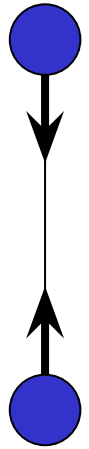
Figure B

Description of the simulation and the Python script

- Define r(0) and v(0) for each atom
  - In the initial configuration the particles are at a distance of at least 0.35. The starting velocities are randomly generated according to the given temperature.
- Iterate over k steps:
  - Calculate the non-bonded forces based on r(i):
    - Initialize force arrays for each atom
    - Calculate forces applied on $atom_j$ due to interaction with $atom_k$
    - Add the force to the array element of $atom_j$
    - Subtract the force from the array element of $atom_k$
  - Integrate to generate new velocities v(i+1) and positions r(i) for each particle
- Total energy of the system ($E_{tot}=E_{pot}+E_{kin}$) should be conserved. Always monitor this during the simulation.

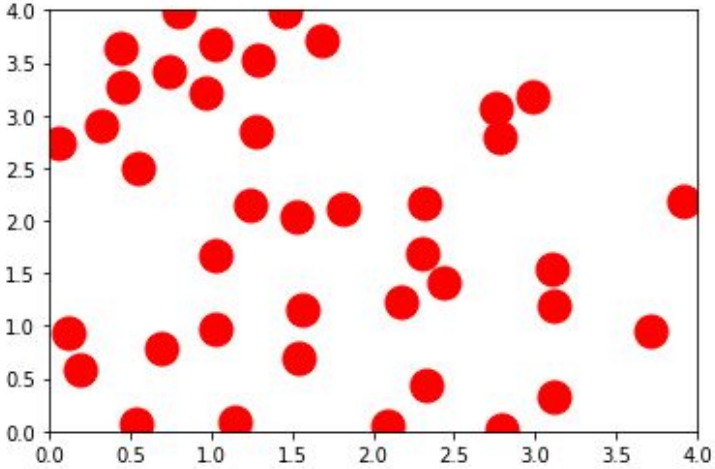
$F_{ij} = - F_{ji}$

(Newton's 3rd law)



First task description

Simulating the dynamic behavior of **40 Argon atoms** in **2 dimensions** for **50 ps** at a temperature of **300 K** in a box with a length of **4 nm** in a periodic boundary condition and a time step of **0.01 ps**.

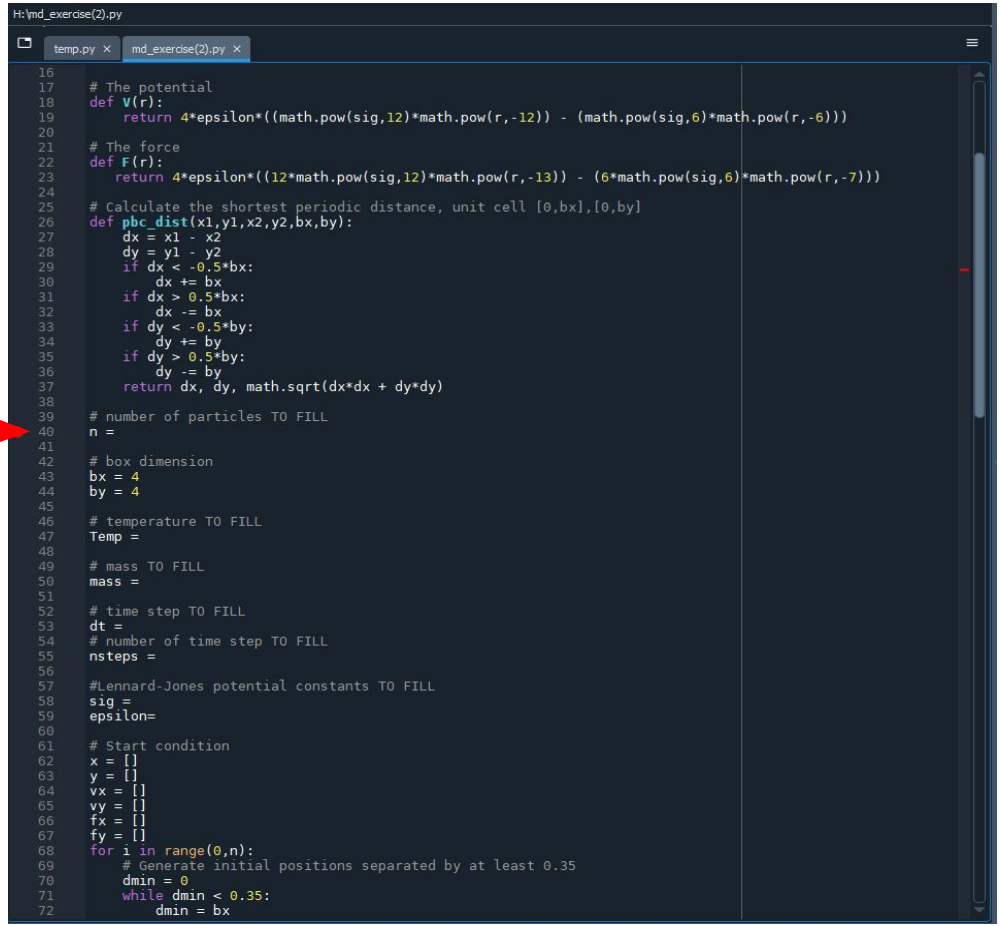


Computer lab – Running simulations

- Complete the missing parts in the python script (#TO FILL) following the instruction. The instruction PDF can be found in Canvas
- Run the script using Spyder
- Address the points in the instruction using the script.
- Answer the questions by presenting your results in a report form. Recall to add your name/names to the report and upload it to canvas as soon as possible and before 23rd of February 2024.

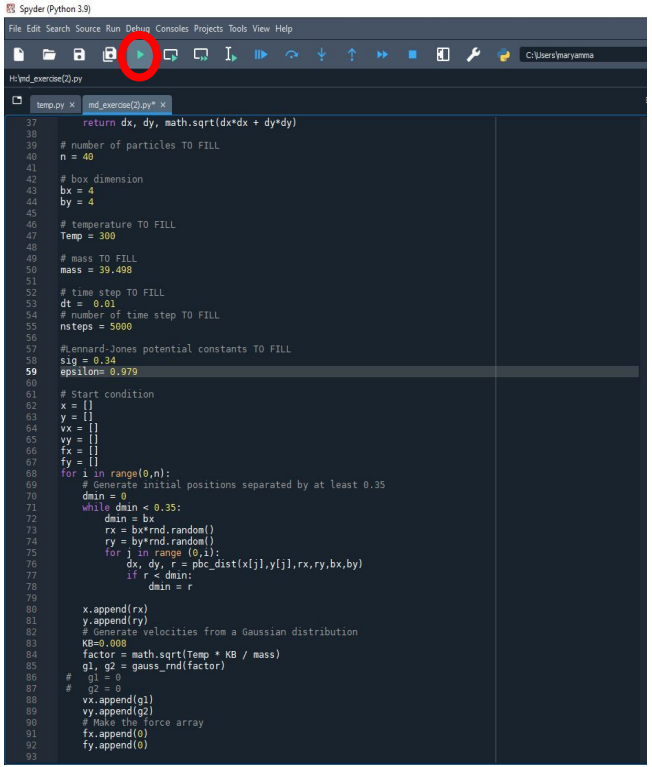
Computer lab – Complete the python script

Fill the needed parameters based on the instruction



Computer lab – Run the script using Spyder

Run the script





Point to address to the report

a) The script uses a simple algorithm to solve Newton's equations of motion for a set of 40 interacting particles.

- Write the expression for the position x(t) and velocity v(t) for the algorithm in the script.
- An alternative algorithm is due to leap-frog. Write the expression for the position x(t) and velocity v(t) in the leap-frog algorithm.
- Discuss possible drawback of the two algorithms

Point to address to the report

b) The atoms interact with van der Waals interactions, which we model using the Lennard-Jones potential. As was explained before, the parameter ε is the well-depth**.** Perform a simulation using a Lennard-Jones potential with a depth of 1.5 kJ/mol (keep the initial σ value) and compare the results with the simulations you have already performed for Argon atoms. In particular:

- Draw the potential energy curve in function of the distance and compare the shapes. Add a plot with both curves in it and explain what has changed.
- Describe the changes in the energies derived from the Python simulations. Support your analysis with suitable graphs.

Point to address to the report

c) Total energy ($E_{tot}=E_{pot}+E_{kin}$) in the simulation should be conserved, so it is good practice to monitor this during the simulation.

- Describe what happens to the total energy if you use a time step larger than 0.01 ps.
- What is the largest time step that one can use to simulate the Argon atoms and keep energy conservation. Support your results with simulation outputs.

Point to address to the report

d) Describe how temperature and mass affect the kinetic energy.

- To achieve this, perform simulations using different values for temperature and mass.
- Tips: change only one value in each simulation.

Point to address to the report

e) Imagine that you want to simulate the conformation rearrangement of a small protein (a system of <100000 atoms) in detail. (the time scale of the process is nanoseconds):

- Which simulation method (Monte-Carlo or Molecular Dynamics) will you use? Why?
- Which molecular model will you use (atomistic or coarse grained model)? Why?

Report's description

- Answer the provided questions and upload the answers as a report (Max 1500 words) before 14th of February 2025 in Canvas.
- Address each point in a separate session.
- Provide all the necessary explanation in a clear way.
- Plots/Figure should have caption, legend and labels on the axis. Values should be written together with unit.

Report corrections:

- Criteria of evalutation
  - Report should be clearly organized
  - All the points are correctly addressed (no wrong statement)
- Procedure
  - Deadline 14th of February
  - Correction within 2 weeks
  - Feedback via Canvas
  - All the discussion concerning the report should take place in canvas (not via email)
  - Resubmission should be done within 1 week from feedback

Questions?

